STUDENT'S T- DISTRIBUTION

Let W and V be independent random variables such that $W \sim N(0,1)$ and $V \sim \chi^2(r)$

The joint pdf of W and V is

$$h(w,v) \begin{cases} \frac{1}{\sqrt{2\pi}} e^{-w^2/2} \cdot \frac{v^{r/2-1} e^{-v/2}}{\Gamma(r/2) 2^{r/2}}, & -\infty < w < \infty \\ & 0 < v < \infty \\ 0, & \text{elsewhere} \end{cases}$$

gamma function

Define $T = \frac{W}{\left(\sqrt{\frac{V}{r}}\right)}$

Define $U = V$

- support set of joint pdf of w & v. $\rightarrow \mathcal{S} = \{(w,v) \mid -\infty < w < \infty, 0 < v, \infty\}$

- support set of joint pdf of t & u. $\rightarrow \mathcal{T} = \{(t,u) \mid -\infty < t < \infty, 0 u < \infty\}$

since $t = \frac{w}{\sqrt{\frac{v}{r}}}$ since $u = v$

- Jacobian $\rightarrow$ $J = \begin{vmatrix} \frac{1}{\sqrt{v/r}} & \frac{-1}{2\sqrt{v^3/r}} \\ 0 & 1 \end{vmatrix}$

$$= \frac{1}{\sqrt{v/r}} = \frac{1}{\sqrt{\frac{u}{r}}} = \sqrt{\frac{r}{u}}$$

$\because u = v$

$$\boxed{|J| = \sqrt{\frac{r}{u}}}$$

$t = \frac{w}{\sqrt{\frac{v}{r}}}$

$\Rightarrow w = t\sqrt{\frac{v}{r}}$

$= t\sqrt{\frac{u}{r}}$

$v = u.$

- The joint pdf of T & U is -

$$g(t,u) = \frac{h\left(t\sqrt{\frac{u}{r}}, u\right)}{|J|}$$

$\frac{1}{|J|}$

$$= \begin{cases} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2 u}{2r}} \cdot \frac{u^{\frac{r}{2}-1} \cdot e^{-u/2}}{\Gamma(\frac{r}{2}) \cdot 2^{r/2}} \cdot \sqrt{\frac{u}{r}}, & -\infty < t < \infty \\ & 0 < u < \infty \\ 0 & , \text{elsewhere} \end{cases}$$

- The marginal pdf of T is -

$$g_1(t) = \int_{-\infty}^{\infty} g(t,u) \cdot du$$

$$= \int_0^{\infty} \frac{u^{\frac{r+1}{2}-1} e^{-\frac{u}{2}\left(\frac{t^2}{r}+1\right)}}{\sqrt{2\pi r} \cdot \Gamma(r/2) 2^{r/2}} \cdot du$$

Substitute $z = \frac{u}{2}\left(\frac{t^2}{r}+1\right) \Rightarrow u = \frac{2z}{\left(\frac{t^2}{r}+1\right)}$

$$= \int_0^{\infty} \frac{(2z)^{\frac{r+1}{2}-1} e^{-z} \cdot 2\,dz}{\left(\frac{t^2}{r}+1\right)^{\frac{r+1}{2}-1} \cdot 2^{\frac{r}{2}+1} \sqrt{\pi}\,\Gamma(\frac{r}{2}) \left(\frac{t^2}{r}+1\right)}$$

$$= \int_0^\infty \frac{z^{\left(\frac{r+1}{2}\right)-1} \, e^{-z}}{\left(\frac{t^2}{r}+1\right)^{\frac{r+1}{2}} \cdot \sqrt{\pi r} \cdot \Gamma\left(\frac{r}{2}\right)} \cdot du$$

$$= \boxed{\frac{\Gamma\left(\frac{r+1}{2}\right)}{\sqrt{\pi r}\,\Gamma\left(\frac{r}{2}\right) \cdot \left(\frac{t^2}{r}+1\right)^{\left(\frac{r+1}{2}\right)}} \quad , -\infty < t < \infty}$$

Already known integral.

$$\boxed{\Gamma\alpha = \int_0^\infty e^{-x} \cdot x^{\alpha-1} \cdot dx}$$

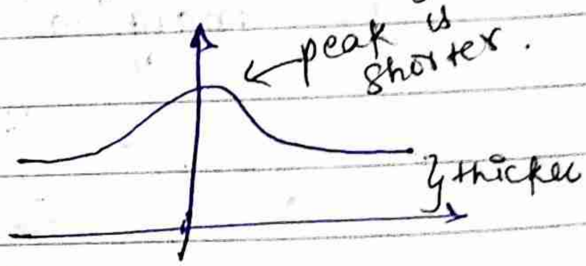
This distribution is called t-distribution.

→ t-distribution depends on r only.

degree of freedom of variable V.

→ and it is symmetric about zero.

• Graph is something like →



Difference in t-distribution graph & normal distribution

* Mean & Variance of t-distribution

$$E(T^K) = E\left(\left(\frac{W}{\sqrt{V/r}}\right)^K\right)$$

$$= E\left(W^K \cdot V^{-K/2} \cdot r^{K/2}\right)$$

$$= r^{K/2}\, E\left(W^K\, V^{-K/2}\right)$$

$$= r^{K/2}\, E(W^K) \cdot E(V^{-K/2})$$ $\because$ W & V are independent

[Page 179]

using this

$X \sim \chi^2(r)$

$$E[X^K] = \frac{2^K \Gamma(\frac{r}{2}+k)}{\Gamma(\frac{r}{2})}, \text{ if } K > -\frac{r}{2}$$

Otherwise does not exist.

$$E(T^K) = \frac{r^{K/2} \cdot E(W^K) \cdot 2^{-K/2} \Gamma(\frac{r}{2}-\frac{k}{2})}{\Gamma(\frac{r}{2})}, \text{ if } \frac{-k}{2} > \frac{-r}{2}$$

$\Rightarrow r > K.$

Otherwise $E(T^K)$ does not exists if $r \leq K$.

- $$E(T) = \frac{r^{1/2} E(W) 2^{-1/2} \Gamma(\frac{r}{2}-\frac{1}{2})}{\Gamma(\frac{r}{2})}, \text{ if } r > 1$$

W has standard normal distribution $W \sim N(0,1)$

$\Rightarrow E(W) = 0$

$$\Rightarrow \boxed{\begin{array}{l} E(T) = 0 \text{ if } r > 1. \\ E(T) \text{ does not exist if } r \leq 1. \end{array}}$$

mean.

- $$E(T^2) = \frac{r^{2/2} \cdot E(W^2) \cdot 2^{-1} \cdot \Gamma(\frac{r}{2}-1)}{\Gamma(\frac{r}{2})}, \text{ if } r > 2$$

$E(W^2) = 1$ since $W \sim N(0,1)$

$$= \frac{r \cdot 2^{-1} \Gamma(\frac{r}{2}-1)}{\Gamma(\frac{r}{2}-1+1)}$$

{Prop $\Rightarrow \Gamma(\alpha+1) = \alpha \Gamma(\alpha)$

$$E(T^2) = \frac{r \cdot 2^{-1} \Gamma(\frac{r}{2}-1)}{\Gamma(\frac{r}{2}-1) \cdot (\frac{r}{2}-1)} = \boxed{\frac{r}{r-2}} \text{ if } r > 2$$

& $E(T^2)$ does not exist if $r \leq 2$

$$\boxed{\begin{aligned} Var(T) &= E(T^2) - (E(T))^2 \\ &= \frac{r}{r-2} \quad \text{if } r > 2 \end{aligned}}$$

$\boxed{\text{\& } Var(T) \text{ does not exist if } r \le 2.}$

Variance.

One Theorem of co-variance & co-relation was left earlier

Co-Variance & Co-relation Coefficients

Theorem) $E(X) = \mu_1 \qquad Var(X) = \sigma_1^2$

$E(Y) = \mu_2 \qquad Var(Y) = \sigma_2^2$

$\rho = \rho(X, Y) =$ correlation coefficient b/w X & Y

If $E(Y/X)$ is linear in X then,

$$\boxed{\begin{aligned} &E\left(\frac{Y}{X}\right) = \mu_2 + \frac{\rho\sigma_2}{\sigma_1}(X - \mu_1) \\ &\text{and } E\left(var\left(\frac{Y}{X}\right)\right) = \sigma_2^2(1-\rho^2) \end{aligned}}$$

Example) $E\left[\frac{Y}{x}\right] = 4x + 3$

$E\left[\frac{X}{y}\right] = \frac{1}{16}y - 3$

$\mu_1, \mu_2, \rho, \frac{\sigma_2}{\sigma_1} = ?$

$E\left[\frac{Y}{x}\right]$ is linear function in x, so

$$E\left[\frac{Y}{x}\right] = \mu_2 + \frac{\rho\sigma_2}{\sigma_1}(x - \mu_1)$$

If $x = \mu_1$, $E\left[\frac{Y}{\mu_1}\right] = \mu_2$

$$4\mu_1 + 3 = \mu_2$$

$$\boxed{4\mu_1 - \mu_2 = -3} \quad -(1)$$

$E\left[\frac{X}{y}\right]$ is linear func. in y, so

$$E\left[\frac{x}{y}\right] = \mu_1 + \frac{\rho\sigma_1}{\sigma_2}(y - \mu_2)$$

If $y = \mu_2$, $E\left[\frac{x}{\mu_2}\right] = \mu_1$

$$\boxed{\frac{\mu_2}{16} - 3 = \mu_1} \quad -(2)$$

Solving (1) + (2)

$$\frac{4\mu_1 + 3}{16} - 3 = \mu_1$$

$$4\mu_1 + 3 - 48 = 16\mu_1$$

$$12\mu_1 = -45 \Rightarrow \mu_1 = -\frac{\cancel{45}\,15}{\cancel{12}\,4} \Rightarrow \boxed{\mu_1 = -\frac{15}{4}}$$

$$\mu_2 = 4\left(-\frac{\cancel{45}^{15}}{\cancel{12}_{3}}\right) + 3 = -12$$

$$\boxed{\mu_2 = -12}$$

Now, $E\left[\frac{Y}{x}\right] = \mu_2 + \frac{\rho\sigma_2}{\sigma_1}(x - \mu_1)$

$$4x + 3 = (-12) + \rho\frac{\sigma_2}{\sigma_1}\left(x + \frac{15}{4}\right)$$

ρ has to be +ve bcz σ are always +ve & for 4 & $\frac{1}{16}$, ρ = +ve

$$\Rightarrow \frac{\rho\sigma_2}{\sigma_1} = 4 \quad -(3)$$

Similarily $\Rightarrow \frac{\rho\sigma_1}{\sigma_2} = \frac{1}{16} \quad -(4)$

(3) × (4)

$$\rho^2 = \frac{1}{4} \Rightarrow \boxed{\rho = \frac{1}{2}}$$

$$\frac{\sigma_2}{\sigma_1} = \frac{4}{1/2} = \boxed{8}$$

F-DISTRIBUTION

$U \sim \chi^2(r_1)$ } Assume that U & V
$V \sim \chi^2(r_2)$ } are independent.

Definition → $f = \dfrac{(U/r_1)}{(V/r_2)} = \left(\dfrac{r_2}{r_1}\right)\left(\dfrac{U}{V}\right)$

Derivation of f-distribution → Reading Exercise.

The pdf of f is called f-distribution and it is given by-

$$g_1(f) = \begin{cases} \dfrac{\Gamma\left(\frac{r_1+r_2}{2}\right)\left(\frac{r_1}{r_2}\right)^{r_1/2}(f)^{\frac{r}{2}-1}}{\Gamma\left(\frac{r_1}{r_2}\right)\Gamma\left(\frac{r_2}{2}\right)\left(1+\frac{r_1 f}{r_2}\right)^{(r_1+r_2)/2}} & , 0 < f < \infty \\ 0 & , \text{elsewhere} \end{cases}$$

* Moments of f-Distribution

$f = \dfrac{(U/r_1)}{(V/r_2)}$, $U \sim \chi^2(r_1)$
$V \sim \chi^2(r_2)$
U & V are independent.

$$E(f^k) = E\left(\left(\frac{r_2}{r_1}\right)^k U^k V^{-k}\right) = \left(\frac{r_2}{r_1}\right)^k E(U^k V^{-k})$$

$$= \left(\frac{r_2}{r_1}\right)^k E(U^k)\, E(V^{-k}) \quad \text{—} (1)$$

Page 179 :- If $X \sim \chi^2(r)$ then

$$E(X^k) = \frac{2^k \Gamma\left(\frac{r}{2}+k\right)}{\Gamma\left(\frac{r}{2}\right)}, \text{ if } k > -\frac{r}{2}$$

Otherwise $E[X^k]$ does not exist.

Using this -

~~E(R K)=~~

since $k > -\frac{r_1}{2}$, always

So, $E[U^k]$ always exists.

$E(V^{-k})$ exists if $-k > \frac{r_2}{2}$, i.e if $r_2 > 2k$

So, $\boxed{E[F^k] \text{ exists if } r_2 > 2k}$

Further simplication is part of Ex 3.6.7.

- for $k=1$,

Eqn ① $\rightarrow$ $E(F)$ exists if $r_2 > 2$
& does not exist if $r_2 \le 2$

$$E(F) = \left(\frac{r_2}{r_1}\right) E(U)\, E(V^{-1}), \text{ if } r_2 > 2$$

$$= \left(\frac{r_2}{r_1}\right) \cdot \frac{2\,\Gamma\left(\frac{r_1}{2}+1\right)}{\Gamma\left(\frac{r_1}{2}\right)} \cdot \frac{2^{-1}\,\Gamma\left(\frac{r_2}{2}-1\right)}{\Gamma\left(\underbrace{\frac{r_2}{2}}_{\frac{r_2}{2}-1+1}\right)}$$

$$= \frac{\left(\frac{r_2}{r_1}\right) \cancel{\Gamma\left(\frac{r_1}{2}\right)} \left(\frac{r_1}{2}\right) \cdot \cancel{\Gamma\left(\frac{r_2}{2}-1\right)}}{\cancel{\Gamma\left(\frac{r_1}{2}\right)} \cdot \left(\frac{r_2}{2}-1\right) \cancel{\Gamma\left(\frac{r_2}{2}-1\right)}}$$

Now use $\Gamma(\alpha+1) = \alpha\Gamma\alpha$

$$= \frac{r_2}{2\left(\frac{r_2-2}{2}\right)} = \boxed{\frac{r_2}{r_2-2}} \quad \text{if } r_2 > 2$$

$E(F)$ does not exist if $r_2 \le 2$

THEOREM 3.6.1 :-

STUDENT'S THEOREM - Let $X_1, \ldots, X_n$ be iid random variables each having normal distribution $N(\mu, \sigma^2)$

Define $\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$

and $S^2 = \frac{1}{(n-1)} \sum_{i=1}^{n} (X_i - \bar{X})^2$

Then,

a) $\bar{X}$ has $N\left(\mu, \frac{\sigma^2}{n}\right)$ distribution.

b) $\bar{X}$ and S^2 are independent.

c) $\frac{(n-1)}{\sigma^2} S^2$ has $\chi^2(n-1)$ distribution.

d) $T = \frac{\bar{X} - \mu}{\left(\frac{S}{\sqrt{n}}\right)}$ has $t(n-1)$ - distribution.

* $Z \sim N(\mu, \sigma^2)$

$\Leftrightarrow \frac{Z - \mu}{\sigma} \sim N(0, 1)$

$$\boxed{\frac{\bar{X}-\mu}{\left(\frac{\sigma}{\sqrt{n}}\right)} \Bigg/ \left(\frac{\sqrt{n-1}\, S}{\sigma} \Big/ \sqrt{n-1}\right) = \frac{\bar{X}-\mu}{(S/\sqrt{n})}}$$

Example) In sixteen, one hour test runs, the gasoline consumption of an engine averaged 16.4 gallons with a standard deviation of 2.1 gallons. Test the claim that the average gasoline consumption is 12 gallons per hour.

Soln)

$n = 16$

$\mu = 12$ ← Claim

$\bar{x} = 16.4$ ← observed

$S = 2.1$

$$t = \frac{\bar{x}-\mu}{\left(\frac{S}{\sqrt{n}}\right)} = \frac{16.4-12}{\left(\frac{2.1}{\sqrt{16}}\right)} = \frac{4.4}{\left(\frac{2.1}{4}\right)}$$

$$= \frac{4.4 \times 4}{2.1} = \boxed{8.38}$$

t- distribution Table

r \ $P(T \le t)$	0.995
1	⋮
2	
⋮	
15	2.947

from t- distribution table,

$P(T \le 2.947) = 0.995$

$P(T > 2.947) = 1 - 0.995$
$= 0.005$

$\therefore P(T = 8.38)$ is extremely negligible

So, it is reasonable to conclude that gasoline consumption exceeds 12 gallons per hour.

Continued after next page.

UNIT - III $\rightarrow$ ELEMENTARY STATISTICAL IN[redacted]FERENCES

STATISTICAL INFERENCE -

Let $X_1, \ldots, X_n$ denote a random sample on a random variable X with pdf (or pmf) $f(x, \theta)$ (or $p(x, \theta)$) where θ is a parameter belonging to a specified set Ω.

Any statistic $T(X_1, \ldots, X_n)$ used to determine θ is called point estimator of θ.

T is said to be unbiased for θ if $E(T) = \theta$

$\bar{X}$ is unbiased estimator for μ.

S^2 is unbiased estimator for σ^2.

$\tilde{T} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$ is a point estimator for σ^2,

but $E(\tilde{T}) \neq \sigma^2$, so it is biased.

$$\tilde{T} = \left(\frac{n-1}{n}\right)S^2$$

$$E(\tilde{T}) = \left(\frac{n-1}{n}\right)\sigma^2 \neq \sigma^2$$

Bias in $\tilde{T} = \sigma^2 - \left(\frac{n-1}{n}\right)\sigma^2 = \frac{\sigma^2}{n}$

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UNIT- II

In continuation of f-distribution & Student's Theorem

Exercise) $X \sim N(\mu_1, \sigma_1^2)$

$X_1, \ldots, X_n$ iid random variables (random sample) on a population with random variable X.

$Y \sim N(\mu_2, \sigma_2^2)$

$Y_1, \ldots, Y_m$ iid random variables on a population with random variable Y.

Sample Variances of Ist & IInd Samples are S_1^2 and S_2^2 respectively.

Then show that $\dfrac{(S_1^2/\sigma_1^2)}{(S_2^2/\sigma_2^2)}$ has $f(n-1, m-1)$ distribution.

Solution) $U \sim \chi^2(r_1)$

$V \sim \chi^2(r_2)$

& (U & V) are independent

$\dfrac{(U/r_1)}{(V/r_2)} \sim f(r_1, r_2)$

$$\left(\frac{(n-1)S_1^2}{(n-1)\sigma_1^2}\right) \Big/ \left(\frac{(m-1)S_2^2}{(m-1)\sigma_2^2}\right) \sim f(n-1, m-1)$$

$$\Rightarrow \frac{(S_1^2/\sigma_1^2)}{(S_2^2/\sigma_2^2)} \sim f(n-1, m-1)$$

UNIT-III continues....

A statistic $T(X_1, \ldots X_n; \theta)$ used to estimate θ is called point estimation of θ.

Parameter space Ω is the set where θ belongs.

T is said to be unbiased estimator of θ if $E(T) = \theta$

* Maximum likelihood Estimator (MLE)

Let $X_1, \ldots, X_n$ be a random sample from a population with random variable X having pdf $f(x;\theta)$ (or pmf $p(x;\theta)$)

The joint pdf of $X_1, \ldots; X_n$ is

$$\prod_{i=1}^{n} f(x_i;\theta)$$

The likelihood function is defined as

$$L(\theta) = \prod_{i=1}^{n} f(x_i, \theta)$$

We want to find a value of θ which maximizes $L(\theta)$.

Such a value if it exists, is denoted by $\hat{\theta}$.

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Example) X is random variable with exponential distribution.

$$f(x;\theta) = \begin{cases} \frac{1}{\theta} e^{-x/\theta} & , x > 0 \\ 0 & , \text{elsewhere} \end{cases}$$

$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta} e^{-x_i/\theta} \quad , x_i > 0$$

to get rid of product rule, we take log.

Since log is increasing function, wherever $\log(L(\theta))$ will be max, $L(\theta)$ will be max at that same pt.

$$\begin{aligned} l(\theta) &= \log(L(\theta)) \\ &= n\log\frac{1}{\theta} + \log\left(e^{-(x_1+\dots+x_n)/\theta}\right) \\ &= -n\log\theta - \frac{1}{\theta}\sum_{i=1}^{n} x_i \end{aligned}$$

- $$\frac{dl(\theta)}{d\theta} = \frac{-n}{\theta} + \frac{1}{\theta^2}\sum_{i=1}^{n} x_i$$

* If $\frac{dl}{d\theta} = 0$, then

$$\frac{-n}{\theta} + \frac{1}{\theta^2}\sum_{i=1}^{n} x_i = 0$$

$$\Rightarrow \theta = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\Rightarrow \hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i = \overline{x}$$

- $$\left.\frac{d^2 l}{d\theta^2}\right|_{\hat{\theta}} = \left.\frac{n}{\theta^2} - \frac{2}{\theta^3}\sum_{i=1}^{n} x_i\right|_{\hat{\theta}}$$

$$= \frac{n}{\hat{\theta}^2} - \frac{2}{\hat{\theta}^3}(n\hat{\theta}) = \frac{n}{\hat{\theta}^2} - \frac{2n}{\hat{\theta}^2} = \frac{-n}{\hat{\theta}^2} < 0$$

$\Rightarrow$ $l(\theta)$ maximizes at $\hat{\theta}$.
& hence $L(\theta)$ also maximizes at $\hat{\theta}$.

$$\hat{\theta} = \frac{1}{n}\sum X_i = \overline{X}$$

Show: $E(\hat{\theta}) = E(\overline{X}) = \theta$

↳ already done in UNIT-II.

$\Rightarrow$ $\hat{\theta}$ is an unbiased estimator of θ.

Example) $X \sim N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

The likelihood function is -

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$l(\mu, \sigma^2) = \log L(\mu, \sigma^2)$$

$$= n\log\frac{1}{\sigma} + n\log\frac{1}{\sqrt{2\pi}} + \left(-\frac{1}{2}\sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)^2\right)$$

$$= -n\log\sigma + n\log\frac{1}{\sqrt{2\pi}} - \frac{1}{2}\sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)^2$$

- $$\frac{\partial l}{\partial \mu} = -\frac{1}{2} \cdot 2 \cdot \sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)\left(\frac{-1}{\sigma}\right)$$

$$= \sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma^2}\right)$$

- $$\frac{\partial l}{\partial \sigma} = -\frac{n}{\sigma} - \frac{1}{2}(-2)\sum_{i=1}^{n}\frac{(x_i-\mu)^2}{\sigma^3}$$

$$\frac{\partial l}{\partial \sigma} = \frac{-n}{\sigma} + \sum_{i=1}^{n} \frac{(x_i-\mu)^2}{\sigma^3}$$

- If $\frac{\partial l}{\partial \mu} = 0$, then

$$\sum_{i=1}^{n} \left(\frac{x_i-\mu}{\sigma^2}\right) = 0$$

$$\Rightarrow \quad n\mu = \sum_{i=1}^{n} x_i \quad \Rightarrow \quad \mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- If $\frac{\partial l}{\partial \sigma} = 0$, then

$$\sum_{i=1}^{n} \frac{(x_i-\mu)^2}{\sigma^3} = \frac{n}{\sigma}$$

$$\Rightarrow \quad \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2$$

$$\Rightarrow \quad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})^2$$

- $$\left.\frac{\partial^2 l}{\partial \mu^2}\right|_{\hat{\mu}} = \sum_{i=1}^{n} \frac{-1}{\sigma^2} = \left.\left(\frac{-n}{\sigma^2}\right)\right|_{\hat{\mu}}$$

$$= \frac{-n}{\sigma^2} < 0$$

- $$\frac{\partial^2 l}{(\partial \sigma)^2} = \frac{n}{\sigma^2} - \frac{3}{\sigma^4}\sum_{i=1}^{n}(x_i-\mu)^2$$

- $$\frac{\partial^2 l}{\partial \mu\, \partial \sigma} = \frac{-2}{\sigma^3}\sum_{i=1}^{n}(x_i-\mu)$$

Diff of $\frac{\partial l}{\partial \mu}$ w.r.t σ $\Rightarrow \frac{d\left(\frac{\partial l}{\partial \mu}\right)}{d\sigma}$

$$\begin{vmatrix} \dfrac{\partial^2 l}{\partial \mu^2} & \dfrac{\partial^2 l}{\partial \mu \partial \sigma} \\ \dfrac{\partial^2 l}{\partial \sigma \partial \mu} & \dfrac{\partial^2 l}{(\partial \sigma)^2} \end{vmatrix} \leftarrow \text{Compute}$$

{ Evaluate this determinate at $(\hat{\mu}, \hat{\sigma}^2)$ and verify that it is positive. }.

If $\dfrac{\partial^2 l}{\partial \mu^2} < 0$ & det. > 0 then $l(\mu, \sigma^2)$ is max at $(\hat{\mu}, \hat{\sigma}^2)$

Second Derivative Test for 2 variables.

After this

MLE of μ is $\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{X}$

So, μ is unbiased estimator

Show: $E(\hat{\mu}) = E(\bar{X}) = \mu$

$\hat{\mu}$ is unbiased estimator of μ.

MLE of $\sigma^2 = \hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$

Show $E(\hat{\sigma}^2) = \left(\frac{n-1}{n}\right)\sigma^2 \neq \sigma^2$

$\hat{\sigma}^2$ is biased estimator of σ^2.

Bias $= E(\hat{\sigma}^2) - \sigma^2$

$= -\dfrac{\sigma^2}{n} \neq 0$

<u>Example)</u> <u>Binomial Distribution</u>

Let $X_1, X_2, \ldots, X_n$ be random sample from a population X with Bernoulli distribution.

Let θ, $0 \leq \theta \leq 1$, be probability of success

$\Omega = [0,1]$. The pmf of X is $p(x;\theta) = \theta^x (1-\theta)^{1-x}$, $x = 0$ or 1.

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$L(\theta) = \prod_{j=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i}$, $x_i = 0$ or 1

$l(\theta) = \log L(\theta)$

$$= \sum_{i=1}^{n} \underbrace{\log(\theta^{x_i})}_{x_i \log\theta} + \underbrace{\sum_{i=1}^{n} \underbrace{\log(1-\theta)^{1-x_i}}_{(1-x_i)\log(1-\theta)}}_{(n-\sum_{i=1}^{n} x_i)\log(1-\theta)}$$

~~for $\theta \in (0,1)$~~

~~$\frac{dl(\theta)}{d\theta} =$~~

for $\theta \in (0,1)$

$$\frac{dl(\theta)}{d\theta} = \frac{\sum_{i=1}^{n} x_i}{\theta} - \frac{(n - \sum_{i=1}^{n} x_i)}{(1-\theta)}$$

- If $\frac{dl}{d\theta} = 0$, then

$$\frac{(1-\theta)\sum_{i=1}^{n} x_i + \theta \sum_{i=1}^{n} x_i}{\theta(1-\theta)} = \frac{n}{(1-\theta)}$$

$$\Rightarrow \sum_{i=1}^{n} x_i = n\theta \Rightarrow \hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- Now comp. $\left.\frac{d^2 l}{d\theta^2}\right|_{\hat{\theta}}$

and show $\left.\frac{d^2 l}{d\theta^2}\right|_{\hat{\theta}} < 0$.

$L(0) = 0 = L(1)$

$\rightarrow$ MLE of θ is $\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i$

$$E(\hat{\theta}) = E\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right)$$

$$= \frac{1}{n}\sum_{i=1}^{n} E(x_i)$$

$$= \frac{1}{n}(n\theta) = \theta$$

Expectation of Bernoulli is θ

$\hat{\theta}$ is unbiased MLE of θ

CONFIDENCE INTERVAL

Let $X_1, \ldots, X_n$ be random ~~variable~~ sample from a population with random variable X, where X has pdf $f(x;\theta)$, $\theta \in \Omega$. Let $0 < \alpha < 1$ be specified. Let $L = L(X_1, \ldots, X_n)$ and $U = U(X_1, \ldots, X_n)$ be two statistics. We say that (L, U) is a $(1-\alpha)100\%$ confidence interval for θ if $P[\theta \in (L, U)] = 1-\alpha$, i.e the probability of (L, U) containing θ is $1-\alpha$.

$(1-\alpha)$ is called the confidence coefficient or confidence level of the interval.

PIVOTAL QUANTITY

Let $X_1, X_2, \ldots X_n$ be a random sample from a distribution with parameter θ which needs to be ~~did~~ determined.

A random variable Q is said to be a pivotal quantity if -

i) Q depends only on $X_1, \ldots, X_n$ and θ;
Q does not depend upon any other unknown parameter.

ii) The probability distribution of Q does not depend upon θ or any other unknown parameter.

Example) Let $X_1, \ldots, X_n$ be a random sample from a normal distribution $N(\theta, 1)$.
Show that $Q_1 = \bar{X} - \theta$ and $Q_2 = \sqrt{n}(\bar{X} - \theta)$ are pivotal quantities.

Tut 13, Ques 1

Solution) $\bar{X} \sim N\left(\theta, \frac{1}{n}\right)$ Using this $X_i \sim N(\mu_i, \sigma_i^2)$, $a_1X_1 + \ldots + a_nX_n \sim N\left(\sum_{i=1}^{n} a_i\mu_i, \ldots\right)$

$Q_1 = (\bar{X} - \theta)$ Using transformation $\left\{ X \sim N(\mu, \sigma^2),\ \frac{X-\mu}{\sigma} \sim N(0,1) \right.$

$\downarrow$ Using transformation

$Q_1 \sim N\left(0, \frac{1}{n}\right)$

$\Rightarrow Q_1$ is a pivotal Quantity.

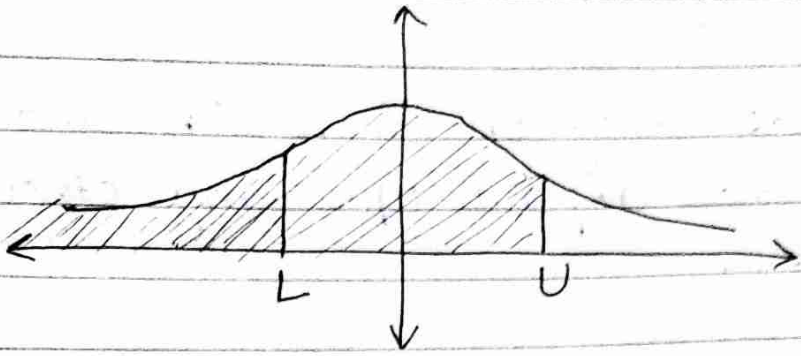
$Q_2 = \sqrt{n}(\bar{X} - \theta) = \sqrt{n}\, Q_1$

$\downarrow$ Using transf.

$Q_2 \sim N(0,1)$

$\Rightarrow Q_2$ is also a pivotal quantity.

* $P[\theta \in (L, U)] = 1 - \alpha$



Let $P[\theta < U] = 1 - \beta$
& $P[\theta \leq L] = \gamma$

$$P[\theta \in (L, U)] = P[\theta < U] - P[\theta \leq L]$$
$$= (1-\beta) - \gamma$$
$$= 1 - (\beta - \gamma)$$

* Let $X_1, \ldots, X_n$ be a random sample from a distribution $N(\mu, \sigma^2)$ where both μ and σ^2 are unknown.

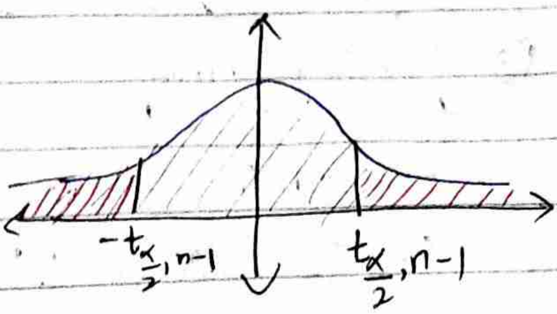
Consider $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$

since σ is also unknown, for pivotal quantity we are using sample variance instead of σ.

Hence, $\frac{\bar{X} - \mu}{S/\sqrt{n}}$ is a pivotal quantity for μ.

Fix $\alpha \in (0, 1)$

$T = \frac{\bar{x} - \mu}{S/\sqrt{n}}$



Take $t_{\frac{\alpha}{2}, n-1} \in \mathbb{R}$ such that

$$P(T \leq t_{\alpha/2, n-1}) = 1 - \frac{\alpha}{2}$$

$$P(T < t_{\frac{\alpha}{2}, n-1}) + P(T \geq t_{\frac{\alpha}{2}, n-1}) = 1$$

$$\Rightarrow P(T \geq t_{\frac{\alpha}{2}, n-1}) = 1 - P(T < t_{\frac{\alpha}{2}, n-1})$$

$$= 1 - \left(1 - \frac{\alpha}{2}\right) = \frac{\alpha}{2}$$ cf. 2nd [illegible] graph

& By symmetry of t-distribution, we get

$$P(T < -t_{\frac{\alpha}{2}, n-1}) = P(T \geq t_{\frac{\alpha}{2}, n-1})$$

$$= \frac{\alpha}{2}$$

$$P(-t_{\frac{\alpha}{2}, n-1} < T < t_{\frac{\alpha}{2}, n-1})$$

$$= P(T < t_{\frac{\alpha}{2}, n-1}) - P(T \leq -t_{\frac{\alpha}{2}, n-1})$$

$$= \left(1 - \frac{\alpha}{2}\right) - \frac{\alpha}{2} = 1 - \alpha$$

$$P\left(-t_{\frac{\alpha}{2}, n-1} < \frac{\bar{X} - \mu}{S/\sqrt{n}} < t_{\frac{\alpha}{2}, n-1}\right) = 1 - \alpha$$

$$= P\left(-\frac{S}{\sqrt{n}} t_{\frac{\alpha}{2}, n-1} < \bar{X} - \mu < \frac{S}{\sqrt{n}} t_{\frac{\alpha}{2}, n-1}\right) \quad \left\{\because \frac{S}{\sqrt{n}} > 0\right\}$$

$$= P\left(-\bar{X} - \frac{S}{\sqrt{n}} t_{\frac{\alpha}{2}, n-1} < -\mu < -\bar{X} + \frac{S}{\sqrt{n}} t_{\frac{\alpha}{2}, n-1}\right)$$

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$$= P\left(\bar{X} - \left(\frac{S}{\sqrt{n}}\right) t_{\frac{\alpha}{2}, n-1} < \mu < \bar{X} + \left(\frac{S}{\sqrt{n}}\right) t_{\frac{\alpha}{2}, n-1}\right)$$

If $\bar{x}$ and s are realized values of $\bar{X}$ and S respectively then,

level of confidence $= 1-\alpha$.

$$\boxed{\begin{array}{l}(1-\alpha)100\% \text{ confidence interval for } \mu \text{ is} \\ \left(\bar{x} - \frac{s}{\sqrt{n}} t_{\frac{\alpha}{2}, n-1},\ \bar{x} + \frac{s}{\sqrt{n}} t_{\frac{\alpha}{2}, n-1}\right)\end{array}}$$

Example) A random sample of 9 observations from a normal population yields the observed statistics: $\bar{x} = 5$ and $\frac{1}{8}\sum_{i=1}^{9}(x_i - \bar{x})^2 = 36$. Determine 95% confidence interval for μ.

$95\% = (1-05)100\%$

$\Rightarrow \alpha = 0.5$

Solution.

$n = 9$

$\bar{x} = 5$

$s^2 = 36 \Rightarrow s = 6$

$$P\left(-t_{\frac{\alpha}{2}, 8} < T < t_{\frac{\alpha}{2}, 8}\right) = 1-\alpha$$

$$P\left(T < t_{\frac{\alpha}{2}, 8}\right) - P\left(T \leq -t_{\frac{\alpha}{2}, 8}\right) = 1-\alpha$$

$$2P\left(T < t_{\frac{\alpha}{2}, 8}\right) - 1 = 1-\alpha$$

$$= 0.95$$

$$\boxed{\begin{array}{l}\text{If } t > 0 \\ P(T \leq -t) \\ = 1 - P(T < t)\end{array}}$$

$\Rightarrow \quad P\left(T < t_{\frac{\alpha}{2},8}\right) = \frac{1.95}{2} = 0.975$

$\Rightarrow \quad P\left(T \le t_{\frac{\alpha}{2},8}\right) = 0.975$ → need to get further value from t distribution table.

Page 710) **t-distribution Table**

Degree of freedom (for t-distribution (n−1))	P(T ≤ t) 0.900	0.950	0.975	0.990 ....
1				
2				
3				
4				
5				
6				
7				
8	1.397	1.860	$\boxed{2.306}$	......
9	1.383	1.833	2.262	......
⋮				

value of t for: $P\left(T \le t_{\frac{\alpha}{2},8}\right) = 0.975$

$\Rightarrow \quad t_{\frac{\alpha}{2},8} = 2.306$

So, 95% confidence interval for μ is

$$\left(\bar{x} - \frac{s}{\sqrt{n}} t_{\frac{\alpha}{2},n-1}, \ \bar{x} + \frac{s}{\sqrt{n}} t_{\frac{\alpha}{2},n-1}\right)$$

$$= \left(5 - \frac{6}{3} \times 2.306, \ 5 + \frac{6}{3} \times 2.306\right)$$

$$= (5 - 4.612, \ 5 + 4.612)$$

$$= \boxed{(0.388, \ 9.612)}$$ Ans

CENTRAL LIMIT THEOREM

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution that has mean μ and finite variance σ^2.
The the distribution function of W_n,

$$W_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$, conveges to Φ, the distribution function of standard normal $N(0,1)$ as $n \to \infty$.

* **Remark-** ~~$Z_n = \bar{X} - \mu$~~

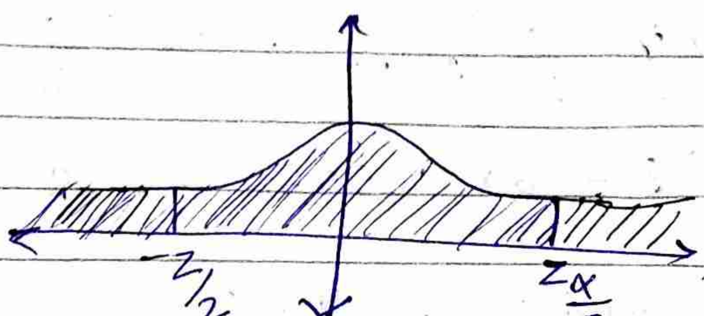
when σ is unknown

The distribution of $Z_n = \dfrac{\bar{X} - \mu}{S/\sqrt{n}}$ is approximately $N(0,1)$.

* **Large sample confidence interval for μ**

$X_1, \ldots, X_n$ random sample with mean μ on a distribution which is not normal

$$Z_n = \frac{\bar{X} - \mu}{S/\sqrt{n}} \rightarrow N(0,1) \text{ as } n \to \infty$$



Let $z_{\frac{\alpha}{2}}$ be a point such that

$$P(Z_n < z_{\frac{\alpha}{2}}) = 1 - \frac{\alpha}{2}$$

$$P(Z_n < z_{\frac{\alpha}{2}}) + P(Z_n \geq z_{\frac{\alpha}{2}}) = 1$$

By symmetry of Normal Distribution $N(0,1)$ we have -

$$P(Z_n < -z_{\frac{\alpha}{2}}) = P(Z_n \geq z_{\frac{\alpha}{2}}) = 1 - P(Z_n < z_{\frac{\alpha}{2}})$$
$$= 1 - \left(1 - \frac{\alpha}{2}\right) = \frac{\alpha}{2}$$

$$P(-z_{\frac{\alpha}{2}}, Z_n < z_{\frac{\alpha}{2}}) = P(Z_n < z_{\frac{\alpha}{2}}) - P(Z_n \leq -z_{\frac{\alpha}{2}})$$
$$= 1 - \frac{\alpha}{2} - \frac{\alpha}{2} = 1 - \alpha$$

$$\boxed{\begin{array}{l}\text{If } z > 0, \text{ then} \\ P(Z_n \leq -z) = 1 - P(Z_n < z)\end{array}}$$

$$1 - \alpha = P\left(-z_{\frac{\alpha}{2}} < \frac{\bar{X} - \mu}{S/\sqrt{n}} < z_{\frac{\alpha}{2}}\right)$$
$$= \dots$$
$$= P\left(\bar{X} - \frac{S}{\sqrt{n}} z_{\frac{\alpha}{2}} < \mu < \bar{X} + \frac{S}{\sqrt{n}} z_{\frac{\alpha}{2}}\right)$$

If $\bar{x}$ and s are realized values of $\bar{X}$ and S, respectively, then $\left(\bar{x} - \frac{s}{\sqrt{n}} z_{\frac{\alpha}{2}}, \bar{x} + \frac{s}{\sqrt{n}} z_{\frac{\alpha}{2}}\right)$ is $(1-\alpha)100\%$ confidence interval for μ.

Example) In sampling from a non normal distribution with a variance of 25, determine the sample size so that length of a 95% confidence interval for μ is 1.96.

Solution

$\sigma^2 = 25 \Rightarrow \sigma = 5$

$1-\alpha = 0.95$

$$P\left(-Z_{\frac{\alpha}{2}} < Z_n < Z_{\frac{\alpha}{2}}\right) = P\left(Z_n < Z_{\frac{\alpha}{2}}\right) - P\left(Z_n \leq -Z_{\frac{\alpha}{2}}\right)$$

$$= P\left(Z_n < Z_{\frac{\alpha}{2}}\right) - \left(1 - P\left(Z_n < Z_{\frac{\alpha}{2}}\right)\right)$$

$$= 2P\left(Z_n < Z_{\frac{\alpha}{2}}\right) - 1 = 0.95$$

$$= P\left(Z_n < Z_{\frac{\alpha}{2}}\right) = 0.975$$

→ have to use normal distribution table to determine value of $Z_{\alpha/2}$.

Page 709) Normal distribution table

here Table has probabilities inside the table.

z	0.00	0.01	....	0.006	0.007	0.008	0.009
0.0	0.500	0.504					
0.1							
⋮							
1.0							
1.1							
⋮							
1.9	0.9713	0.9719	...	$\boxed{0.9750}$	0.9756	...	
2.0				0.9803			

By normal distribution table,

$$Z_{\frac{\alpha}{2}} = 1.9 + 0.06 = 1.96$$

Length of confidence interval $= \frac{2\sigma}{\sqrt{n}} \cdot Z_{\alpha/2}$

$$1.96 = 2\left(\frac{5}{\sqrt{n}}\right) 1.96$$

$$\sqrt{n} = 10$$

$$\boxed{n = 100}$$ Ans.

NOTE → When sample size is small & variance is not given, we use t- distribution.

small sample size $= (n < 30)$. conventionally